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(See 6.5)

MILITARY SPECIFICATION

MOCK-UPS FOR SHIPBOARD COMMAND AND CONTROL, COMBAT SYSTEMS, AND COMMUNICATIONS STATIONS AND SPACES

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the development of mock-ups and models for shipboard command and control, combat systems, and communications stations and spaces for all surface and subsurface combatant ships, amphibious ships, auxiliaries, special purpose vessels and crafts for new construction as well as for backfit a conversion and modernization. It does not address requirements for the mock-up of other shipboard spaces although the requirements specified herein are generally applicable to any shipboard space, station or area. Operational interface requirements and the effect on operating personnel shall be given prime consideration throughout mock-up development. The ultimate objective is to achieve the optimum feasible integrated physical arrangement of personnel, equipment, systems, furniture, fixtures and other installed facilities (see 6.1), with particular attention to:

(a) Operational concepts of the station or space (functions to be performed).
(b) Functional adjacency requirements (equipment and operator positions).
(c) Safety features (to personnel and equipment).
(d) Operating personnel habitability.
(e) Optimum arrangement of equipment, control devices, information displays, exterior and interior communications and other operating facilities.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(f) Operating visibility and other human factor requirements.
(g) Equipment environmental conditions.
(h) Accessibility for ease of maintenance or repair.
(i) Methods of removing equipment from spaces for maintenance or replacement.
(j) Elimination or consideration of interferences.
(k) Where required, mandatory physical security requirements.
(l) Interrelationships of other stations and spaces.

1.2 Classification.

1.2.1 Types. Mock-ups shall be of the following types, as specified (see 6.2.1):

Type I - Reduced scale mock-up.
Type II(O) - Full scale operational mock-up.
Type II(NO) - Full scale non-operational mock-up.
Type II(F) - Full scale functional mock-up.
Type III - Small scale mock-up.

1.2.1.1 Type definitions.

1.2.1.1.1 Type I. A reduced scale mock-up, which may be larger or smaller than 1/8 scale, if so specified, but smaller than full scale, primarily intended for use by the shipbuilder or a contractor for structural design considerations of a shipboard space or station, or for area arrangement considerations (such as partial mock-up of one section of a bulkhead or a mock-up of a flight deck arrangement).

1.2.1.1.2 Type II(O). A full scale exact replica of a shipboard space or station, inclusive of actual operable equipment, furniture, and support systems (such as electrical power, cooling, interior communications, lighting) required to demonstrate the installed equipment and operator interfaces within an at-sea operational environment. Equipment, furniture, and systems not required for demonstration purposes (such as overhead ducting and cable runs) are simulated within the mock-up through modeling. The primary purpose of this type mock-up, normally constructed at the termination of contract design and at a land based test site (LBTS), is to assure the integration of the software and hardware into an operational combat system. A secondary purpose is to serve as a training aid for training operational personnel.

1.2.1.1.3 Type II(NO). A full scale exact replica of a shipboard space or station similar to the type II(O) mock-up, with the exception that there is no requirement for actual operable equipment and equipment support systems (such as power and cooling) and is, therefore, non-operational. To the extent feasible and practical, however, actual non-operational equipment and equipment enclosures (such as consoles, communications hardware, furniture, plotting boards, indicators, CCTV monitors, and so forth) should be included within the type II(NO) mock-ups to facilitate verification of installation design and ship and Government furnished equipment (GFE) interfaces (see 3.3.4.1.) The type II(NO) mock-up is constructed by the shipbuilder, or contractor, in order to provide for a more thorough evaluation of space or station equipment arrangements, operator positions, human factor engineering consideration and physical constraints than would be available through the construction of either a type I

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and type III mock-up, when system operability is not a prime consideration. Type II(NO) mock-ups, developed by the shipbuilder during detail design, form the basis for configuration control. Within the scope of the type II(NO) mock-ups exist such special purpose mock-ups as for submarine Rapid Installation Plan (RIP) requirements. When required of a contractor by cognizant authority, such amplifying details as may be required will be specified for construction of the mocks-ups.

1.2.1.1.4 Type II(F). A purely functional mock-up of actual shipboard equipment, exclusive of physical boundaries, located on a convenient platform, connected, and capable of system operation. The primary purpose of the type II(F) mock-up is for the verification and check-out of total system cabling design requirements and for operational testing of equipment and computer programs. Wireway routing and cable separation will conform to the Naval Sea Systems Command (NAVSEA) 0967-LP-283-5010 to the extent specified by NAVSEA. The type II(F) mock-up is applicable when equipment demonstration is required and when a type II(O) mock-up will not be available.

1.2.1.1.5 Type III. A small scale (1-1/2 inches equals 1 foot, or 1/8 scale) non-operational mock-up, employed as the basis for the preparation of contract arrangement drawings, and for design validation of operational space and station arrangements which do not require equipment demonstration, nor represent significant human factor engineering considerations, nor incorporate critical constraints due to arrangement complexity, operational and service area requirements and physical obstructions.

1.2.2 Classes. Models of equipment components, systems, furniture, fixtures and other installed material constructed as part of the mock-up or for the purpose of individual evaluation shall be of the following classes (see 6.2.1):

Class 1 - Exact reproduction of the shape and external dimension of an item.
Class 2 - Reproduction of the general shape and major external dimensions.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and publications. Unless otherwise specified, the following specifications, standards and publications of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-O-24312 - Operational Stations Book (OSB).

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STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

PUBLICATIONS

Naval Sea Systems Command (NAVSEA)

0900-LP-098-7010 - Shipboard Command and Control Systems and Associated Configurations, Manual for Design and Development of.

(Copies of specifications, standards and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code (DoD adopted)

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT

National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 1616 "P" Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1. General. Unless otherwise specified, mock-ups shall be constructed to show space perimeters (such as false decks, overheads, portable and structural bulkheads), shock mounts and foundations, space requirements for air conditioning, piping, lighting, insulation, wireways, and all obstructions or protrusions (such as ladders, beams, girders and stanchions) which are located in the mocked-up space or station. Each mock-up shall be detailed so as to identify the functions to be performed within the mocked-up space or station. Mock-up construction shall be of a simplicity which permits easy manufacture.

3.2 Design.

3.2.1 Guidance. Specific spaces or stations to be mocked-up shall be as specified by NAVSEA (see 6.2.1). These spaces or stations, in keeping with the Chief of Naval Operations (CNO) general policy requirements (especially for new construction), will normally be inclusive of such command and control, combat systems, and communications spaces as the combat information center (CIC), navigation bridge area (pilot house, bridge wings and chart room), primary flight control station, flag command and other operational control centers, weapons control, main communications spaces, antisubmarine warfare module, sonar control and such other spaces or stations as may be required. Space or station mock-ups for submarines shall normally include the attack center, sonar control room and radio ECM room. Mock-ups shall be developed in accordance with direction and guidance furnished by NAVSEA. Drawings, specifications, schedule of equipment, design work study (DWS) results, or other guidance may be provided, but the mock-up design shall be such that the ship structure, ventilation, lighting, insulation, coolant piping, wireway requirements, other interferences, and all items of equipment (including mounting facilities) will be considered in the actual mock-up arrangement. Type II mockups shall be designed using arrangement drawings developed by the ship design activity contractor (see 3.2.2, 3.2.4, 3.3.6.1 and 6.3).

3.2.2 Contractor deviations. Contractor deviations from NAVSEA furnished direction and guidance shall require NAVSEA approval. When submitting his request for approval, the contractor shall cite the specific departures from NAVSEA furnished direction and guidance. The contractor shall describe departures in detail and shall indicate the reasons for these departures.

3.2.3 Authorized changes. Authorized changes to a mock-up will be issued by NAVSEA. An authorized change to a mock-up may vary from the original NAVSEA furnished direction and guidance for the ship and applies only to the mock-up. Changes to the ship or to contract documentation applicable to the ship (such as contract or contract guidance drawings, or ship specifications) are not automatically authorized as the result of an authorized change to a mock-up.

3.2.3.1 During presentation. Rearrangement of mock-ups resulting from NAVSEA sponsored presentations shall be incorporated into the mock-up and into mock-up documentation (see 3.4.5.1) upon authorization of NAVSEA.

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3.2.4 Space relationships. Where practicable, spaces mocked-up shall be laid out in the same relationship as the installation aboard the applicable ship. Guidance for shipboard compartmentation and passageway configuration, space allocation and major structural interferences (such as ladder wells, fan rooms, access and egress clearances) will be provided by NAVSEA.

3.2.4.1 Adjacent areas. Spaces located immediately adjacent to a mocked-up space shall be outlined on the base of the mock-up to show adjacent space relationships. If the adjacency is of particular importance to the operations and traffic flow of the basic space being mocked-up, the adjacent compartment shall be roughed in (bulkheads, doorways, deck) but not furnished or equipped, unless directed by NAVSEA.

3.2.4.2 Bridge wings. Where NAVSEA specifies that primary conning stations shall be mocked-up, at least one bridge wing shall also be mocked up. Where bridge wings are configured differently, both shall be included in the mock-up.

3.2.4.3 Control bridges. Where the requirement is to mock-up a control space (such as flight or helicopter control, exposed conning station, replenishment stations, visual communications stations, debarkation control or flag control stations), associated bridges shall be included in the mock-up of the control station or space.

3.3 Construction.

3.3.1 Safety precautions.

3.3.1.1 Electric wiring used within mock-ups for temporary lighting purposes shall be in accordance with the National Fire Protection Association NFPA 70 and other industrial regulations.

3.3.1.2 Plastic and composition material used in the construction of a mock-up shall be of fire-retardant and non-toxic fume types.

3.3.1.3 Combustible materials, such as plywood, wood and fiberboard, used in the construction of type II mock-ups, shall be painted with fire retardant paint or impregnated with fire retardent material. Type II mock-ups shall be constructed within spaces, or at sites, which are equipped with approved fire prevention and protection measures in accordance with the local fire regulations for the area in which the mock-up is constructed. Portable fire extinguishers, and other portable fire fighting equipment, required in the space arrangement shall be installed within the mock-up and shall be in working order. In addition to the above requirements, type II mock-ups which contain actual shipboard equipment of high value (such as NTDS and radar consoles, navigation and radio) shall be equipped with a UL approved smoke and or high temperature detection system, meeting National Fire Protection Association, NFPA 70 standards and capable of actuating remote audio and visual alarms at guard stations. Further, such mock-ups shall be additionally equipped with a total flooding portable extinguishing system (such as HALON) installed at selected locations on the exterior of the mock-up and projecting into the mocked-up space over high value equipment, or as a total flooding system.

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3.3.2 Size and scale.

3.3.2.1 Types I and III mock-ups. Type I mock-ups shall be constructed to any scale less than or greater than 1/8 scale, but less than full scale, as specified. Unless otherwise specified (see 6.2.1), type III mock-ups shall be constructed to a scale of 1-1/2 inches equals 1 foot or 1/8 scale. The overall size of each section of type I and type III mock-ups shall be such to permit their passage through a 6 foot 8 inch high by 32-inch wide doorway; however, no dimensions shall exceed 120 inches.

3.3.2.2 Type II. Type II mock-ups, with the exception of type II(F) mock-ups, shall be constructed to actual size (full scale).

3.3.3 Visibility. Except as otherwise specified herein, overheads and overhead attachments (such as, beams, girders, piping, ducting, wireways, fiddle boards, handsets, telephone jackboxes and speakers) shall be clearly visible in mock-ups.

3.3.3.1 Type I mock-ups. In a type I mock-up, overheads may be simulated by the use of a transparent material. Overhead attachments shall be shown as a removable unit or subunit. Openings representing partial removal of plating from bulkheads shall be used to the degree necessary to provide visibility to all parts of the mock-up and accessibility during construction or modification.

3.3.3.2 Type II mock-ups. A type II mock-up, as a full scale replica of the space or station represented, shall accurately display all hull fittings, access plates, escape scuttles, beams, girders, stanchions, ventilation ducting, piping, wireways, overhead attachments and other obstructions and protrusions. Provisions shall also be included in bulkheads and hull sections for visibility behind all installed units which have rear cabling, piping or other service requirements. For submarine RIP mock-ups, foundations, all critical dimensions, such as required installation and access area dimensions shall be identified in the mock-up.

3.3.3.3 Type III mock-ups. Overhead decking, beams, girders, piping and ducting shall not be shown in a type III mock-up unless they present a known structural interference. Overhead attachments associated with manual operation of a type III mocked-up space (such as fiddle boards, speakers, indicators, handsets and plotting boards) shall be shown.

3.3.4 Materials. Materials used in the construction of mock-ups and models shall be such to achieve maximum economy in construction and maintenance. The structural envelope of the mock-up shall be constructed of durable material with sufficient strength and rigidity in order to support attachments, prevent distortion and provide protection. For types I and III mock-ups, for which shipment is required, the structural envelope shall be of sufficient strength to enable transportation of the mock-up. The base of the mock-up shall be sufficiently strong to support the structure. Material requirements shall be as specified herein for the applicable type of mock-up:

(a) Type I mock-up compartment envelopes may be constructed of wood, plastic, metal or composition material. Models of fixed equipment contained in a type I mock-up shall be made of light-weight wood, plastic or metal, and shall be held in place by

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nails, screws, glue, tacks, magnets and so forth. Where magnets are used, they shall be of sufficient strength to hold the representative equipment in place on the mock-up's overhead, deck or bulkhead, as appropriate. Furniture models shall not be fixed within the mock-up.

(b) Type II mock-up compartment envelopes and models shall be constructed of wood, metal or composition material. When constructed by a contractor other than the ultimate user (the shipbuilder), the mock-up shall be constructed in sections, suitable for shipping by common carrier (see 5.1.1), and joined together with screws or bolts which can be easily removed for dismantling.

(c) Type III mock-up compartment envelopes shall be constructed of heavy cardboard, lightweight wood, composition material, plastic or other suitable material. Models of fixed equipment shall be made of lightweight wood, plastic or metal, and held in place by non-permanent glue. When a requirement for shipment outside of the local area or overseas is specified, type III mock-ups shall be joined with screws or nails to minimize structural distortion and shall be constructed in such a manner as to be suitable for shipping by a common carrier (see 5.1.1).

3.3.4.1 Actual equipment. Actual equipment, such as Naval Tactical Data Systems (NTDS) consoles, radar displays, navigation and other electronic equipment, connectors, mounting hardware, piping, furniture and fixtures, ultimately to be installed in the shipboard space, and which is available and approved for use by NAVSEA, shall be utilized to the maximum extent practicable to fulfill mock-up purposes and to add realism in type II mock-ups. The intended use of actual equipment, inclusive of equipment weights, must be identified to the contractor prior to a request for bid, to assure adequate supporting structures for the units (see 6.2.1).

3.3.5 Painting. Type I and II mock-ups and their associated models shall have at least one coat of paint, colored as proposed for the actual shipboard installation, if practical. Actual shipboard equipment utilized for a type II mock-up shall not be repainted. Type I or type III mock-ups, utilized to display partial rearrangement equipment installation requirements for backfit, modernization, or conversion purposes, may use distinctively painted proposed new equipment models so that they are readily distinguishable from other models in the mock-up. Structures necessary to support the mock-up, that are not representative of the space being mocked-up, shall be painted a distinguishing color.

3.3.6 Models.

3.3.6.1 Construction guidance. For GFE, NAVSEA will provide standard drawings or other guidance for the contractor's use in constructing models of equipment components, units of equipment, items of furniture or fixtures. Contractor's guidance shall be used to construct models of commerically supplied items. Models of constructional interferences (such as ventilation ducts, piping, insulation, structural members and wireways) shall be developed by the contractor using Navy approved documentation. For requirements concerning the construction of personnel models see 3.3.8. For full scale GFE or commercially supplied items see 3.3.4.1.

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3.3.6.2 Equipment. Except as otherwise specified hereinafter, equipment models shall be of class 1 construction and may include actual equipment enclosures where verification of installation interfaces (such as cabling, connectors and piping) are required.

3.3.6.2.1 Identification. Each item of equipment in a type II mock-up shall be identified. The equipment model shall be labeled with military designation and the equipment nomenclature. Components may be identified by a small identification plate. For type I and type III mockups, equipment identification shall be provided in either the Mock-up Booklet or space arrangement drawing, or both (see 3.4).

3.3.6.2.2 Functional details. Equipment functional components, parts and devices (such as, switches, scope faces, connector jacks, meters and pushbuttons) shall be shown and labeled as to identify their function on models displayed in type II mock-ups. These devices shall be shown by means of the use of actual equipments (see 3.3.4.1), or by means of affixing to the model photographic enlargements, facsimiles or decals. Details (such as, meter divisions and scope face markings) need not be shown. Equipment with a large number of operational switches (such as, action cutout, sound powered or radio selector switches) shall be labeled with each dial or switch (position) nomenclature and circuit designation.

3.3.6.2.3 Access requirements. The space required to fully extend equipment or furniture drawers, equipment chassis, console compartment doors and space required for maintenance or replacement of equipment, including requisite working space for maintenance personnel, shall be provided within the space mocked-up as identified within the cognizant space arrangement drawing.

3.3.6.2.4 Operable equipment. Except as specified in 3.3.7.1, and for those models specifically constructed as working models (whether incorporated within a mock-up or not), equipment displayed in a mock-up need not be electrically operable.

3.3.6.2.5 Plotting or display boards. Plotting and display boards shall be of class 2 construction. Where the actual equipment (see 3.3.4.1) is not available for installation within a type II mock-up, models of plotting or display board plotting surfaces shall be constructed of a transparent material, enabling inspectors to view models of installations located behind the board (see 3.3.7.1).

3.3.6.2.6 Foundations and braces. Within a type II mock-up, foundations, equipment braces, racks and environmental cabinets shall be of class 2 construction. Detail of foundations (such as spacers, shims or slotted holes), shock mounts, and obstructions to correctly depict actual heights and clearances shall be provided. Foundations and braces are not required for type I or type III mock-ups, unless they constitute or are an integral part of the equipment or are of significant dimension to warrant display. Equipment racks (such as communications equipment racks) and environmental cabinets shall be mocked-up for type I and type III mock-ups.

3.3.6.2.7 Electrical. Except as specified in 3.3.7.1, electrical fixtures (such as switchboards, distribution panels, power disconnect switches and lighting fixtures) shall be of class 2 construction to demonstrate space requirements.

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3.3.6.2.8 Cables. Transmission lines (cables, waveguides and trunks), wireways, shielding, adapters, junction boxes and connectors shall be of class 2 construction, except for type II(O) and submarine RIP mock-ups, which shall be class 1 construction.

3.3.6.3 Furniture. Furniture and fixtures shall be of class 1 construction. For type II mock-ups, chairs, benches and similar functional items of furniture shall be constructed with sufficient strength to support personnel. Actual chairs, benches, desks and similar functional items of furniture shall be used, if available, or if specified.

3.3.6.3.1 Identification. Items of furniture and fixtures (such as chairs, desks, tables, plotting and display boards, shelves and clocks) shall be identified as specified for equipment in 3.3.6.2.1. Items readily identified by shape (such as different types of chairs) need not be labeled with the item's nomenclature.

3.3.6.3.2 Access requirements. For access requirements, see 3.3.6.2.3.

3.3.6.4 Hull structures. Hull structures shall be of class 2 construction. For type II mock-ups the thickness of bulkhead and overhead insulation shall be included, or simulated, within the structures when equipment is located against or on bulkheads, and or where access clearance requirements are required.

3.3.6.4.1 Space identification. Each mocked-up space and spaces immediately adjacent to a mocked-up space shall be identified and labeled. Functional areas within the mocked-up space shall also be identified and labeled with functional names. For purposes of orientation, shipboard locations (such as ship centerline and frame numbers) shall be clearly visible when viewing a mock-up. For additional requirements see 3.3.6.4.2 and 3.3.6.5.

3.3.6.4.2 Apertures. Models of doors, hatches, scuttles, portholes and windows (except pilot house and flight control windows) for type I and type III mock-ups shall be fabricated and fastened for ease of relocation during NAVSEA sponsored inspections. Doors to mocked-up spaces requiring physical security protection shall be identified as to door type - secure or non-secure. In type II mock-ups, doors, hatches, scuttles, portholes and windows shall be operable as necessary to demonstrate adequate clearance for opening and closing and for operating handwheels, doorknobs and locks.

3.3.6.4.3 Walkways. Walkways (such as decks, platforms and ladders) and similar items for type II mock-ups shall be constructed of material of sufficient strength to support the station's personnel requirements.

3.3.6.4.4 False decks. In spaces to be mocked-up which have more than one deck level, false decking shall be shown.

3.3.6.5 Message passing facilities. Message passing facilities (such as passing windows, scuttles and tubes) shall be shown of class 2 construction. For type I and type III mock-ups, these items shall meet the relocation requirements specified in 3.3.6.4.2. Passing window requiring physical security protection shall be identified as specified for doors in 3.3.6.4.2.

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3.3.6.6 Piping, valves and test fixtures. In type I and II mock-ups, piping of size 1-1/2 inch nominal pipe size and above, valves and test fixtures shall be of class 2 construction. Pipe hangers shall be in the proper locations. In type II submarine space mock-ups except type II(O), all piping, fittings, valves and tubing (regardless of size) shall be of class 1 construction. Insulation shall be shown as necessary.

3.3.6.7 Air conditioning and ventilation. Air conditioning and ventilation systems shall be in type I and type II mock-ups, inclusive of insulation and flanges, shall be of class 2 construction. Type III mock-ups shall include ventilation or ventilation ducting, class 2 construction, when the location of such ducting is known and presents an obstruction to equipment installation.

3.3.7 Lighting requirements. Lighting shall be adequate for the inspection of a mock-up and the safety of inspection personnel.

3.3.7.1 Type II mock-ups. For type II mock-ups, detail operational and emergency lighting shall be installed in order to determine the suitability of operational lighting conditions. Edge lighting for actual plotting and display boards shall be operable. If the actual boards are not installed, their models shall be lighted as realistically as possible. Actual switches and switchboards shall be internally or externally lighted as appropriate. Externally lighted switches and switchboards (actual and models) shall be lighted by means of overhead spotlights. Models of internally lighted switches and switchboards shall be labeled to indicate that the device is internally lighted. Where possible, the actual lighting intended for general and maintenance purposes shall be installed and shall be operable. However, the installation of general and maintenance lighting, other than lighting required to meet the requirements specified in 3.3.7, should not delay the construction of the mock-up.

3.3.8 Personnel. In type III mock-ups, 1/8 scale personnel models or dolls of class 1 construction shall be positioned at key operator positions to provide the mock-up realism, to enable a more refined evaluation of physical constraints within the mocked-up space and at operator stations, and to enable the evaluation of human engineering aspects associated with operator positions, consoles (such as operator sit-stand capability, reach, twist, bend). Personnel models or dolls shall be three-dimensional, representative of the 90-percentile man, and have movable arms and legs, capable of either sit or stand positioning, or both. Personnel models or dolls may either be manufactured from light wood or obtained from commercial sources (such as toy stores or hobby shops).

3.4 Presentation. Mock-ups shall be presented to demonstrate design feasibility and for inspection of operational suitability. Mock-up presentations will be sponsored by NAVSEA. Persons selected for presentation of the mock-up shall be intimately familiar with the detailed design, the design rationale, and its operational application. For presentations involving contractor constructed mock-ups, the contractor shall have available at the presentation site engineers knowledgeable in the detailed design, its development, and associated design rationale. Formal mock-up presentations shall be accompanied by a mock-up booklet (see 3.6). Additionally, the space arrangement drawing of the space mocked-up shall be displayed. For type II submarine RIP mock-ups, a demonstration of the equipment rapid installation using the load path drawings shall be successfully accomplished.

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3.4.1 Place of presentation. Mock-ups shall be presented at a place to be specified by NAVSEA (see 6.2.1). Normally, type I and type III mock-ups are presented within the greater Washington, DC area; type II mock-ups are normally presented at the contractor's facility. Type II(NO) mock-ups may be presented within the geographical area, for review convenience and if space is available, and shipped to the contractor's facility upon completion of inspection, if directed by NAVSEA.

3.4.2 Schedule.

3.4.2.1 Mock-up availability. Type I mock-ups shall be available for presentation at or before the 40 percent point of shipbuilding contract period (detail design), type II at or before the 50 percent point, except for type II mock-ups developed by other than the shipbuilder for contract design purposes and for submarines where the availability shall be at least 30 months prior to launch of the ship. Type III mock-ups are normally constructed and presented during ship design, prior to completion of contract drawings. Type III mock-ups shall also normally be constructed, when required, to assist in the redesign of complex spaces or stations, or the installation of equipment within the space, for backfit and modernization and conversion. In the case of type I and type II mock-ups, the mock-up availability shall be such so as not to delay the production of ship construction drawings or construction of the ship (see 3.2.1 and 3.4.2.4). (The actual schedule for mock-up availability may be different from the above, in accordance with specific contract provisions.)

3.4.2.2 Mock-up construction plan. When required by NAVSEA, the contractor shall provide a design, construction, inspection and presentation schedule with milestones, and it shall update at each status review all required dates for GFE, and information shall be listed in accordance with the data ordering document specified (see 6.2.2).

3.4.2.3 Inspection schedule. Inspection of mock-ups normally will be conducted in the following three parts as specified:

(a) Part one - Working level technical review by cognizant NAVSEA engineering codes and supporting systems engineering activities (such as, NAVELEX).
(b) Part two - Preliminary formal inspection by cognizant technical codes (NAVSEA and supporting systems engineering activities), Fleet representatives, and CNO Program Sponsor representatives.
(c) Part three - Formal inspection by the CNO or CNO Program Sponsor representatives.

Working level inspections may be scheduled before all cabling, small components, and small piping features are represented; however, for the part two and three inspections, the mock-up and mock-up booklet shall be complete. Re-inspections may be required at various points subsequent to the initial inspection and prior to the formal inspection, in order to monitor corrective actions taken to remedy previously noted discrepancies.

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3.4.2.4 Government notification. The contractor shall notify NAVSEA of a mock-up availability for presentation. Notification shall allow NAVSEA approximately 3 months for scheduling presentations. The availability shall be at such time to allow approximately 2 additional months for approval after presentation is scheduled (see 3.4.2.1).

3.4.3 Contractor-provided services. Where the contract specifies that the mock-up shall be presented at the contractor's facility (see 6.2.1), the contractor shall provide the inspection party, during NAVSEA sponsored scheduled inspections and presentations, services as specified below:

(a) Have available at the mock-up site personnel technically qualified to answer questions on the mock-up and the equipment and systems reflected therein (see 3.4).
(b) Have available at the mock-up site copies of the applicable NAVSEA furnished arrangement drawings, systems drawings and other applicable design documentation.
(c) Provide clerical assistance for taking notes during the formal presentation and for preparation of inspection reports inclusive of a listing of attendees, as required.
(d) Provide a conference room in the vicinity of the mock-up site.
(e) Have a mock-up booklet available at the mock-up site (see 3.6).

3.4.4 Authorized changes. For authorized changes made during a NAVSEA sponsored presentation see 3.2.3 and 3.2.3.1.

3.4.5 Final approval. Mock-up final inspections shall be scheduled and presented as specified in this specification. Approval of a mock-up arrangement and associated documentation will be issued by NAVSEA.

3.4.5.1 Technical data. Within 30 working days, or as specified, following final mock-up approval, the contractor shall update the mock-up to conform to the approved arrangement and shall prepare the following documentation as specified in the data ordering document (see 6.2.2):

(a) A list of all proposed ship specification changes required by the approved mock-up configuration.
(b) A reproducible and five prints of arrangement drawings which represent the approved NAVSEA and CNO configuration.
(c) Photographs of the approved mock-up. The number of copies of each photograph to be determined by NAVSEA.
(d) Fifteen copies of the mock-up booklet (see 3.6.1), corrected to reflect the approved mock-up.

3.5 Photographs.

3.5.1 Photographic details. Photographs shall be taken to record all angles of a mock-up. Equipment, equipment components, items of furniture or fixtures, structural interferences and ancillary devices displayed in a mock-up shall be visible within a set of mock-up photographs. If at all practical, photographic angles shall be such to show equipment scope faces, dials and switches as specified in 3.3.6.2.2. Each photographic print shall be labeled to

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show: ship or ship class, direction of view, space or compartment, functional area, operator positions, major equipment and facilities, shipyard or command providing photographic services; and date taken. The contractor may retain one print for his files. Photographs shall be black and white, unless color photographs are specified. When specified (see 6.2.1), distribution of negatives and prints will be as specified by NAVSEA.

3.6 Mock-up documentation. At least 2 weeks prior to a scheduled formal NAVSEA mock-up presentation, the contractor shall prepare and have available for distribution to the inspection team members and for reference at the mock-up site a mock-up booklet (see 3.6.1) and, where specified by NAVSEA, an outline operational stations book (OSB(O)) (see 3.6.2 and 6.2.1).

3.6.1 Mock-up booklet. The mock-up booklet shall be subtitled "Detailed Design and Rationale" and identify spaces included (such as Command and Control Combat Systems, Communications Space and Replenishment Stations) and shall be unclassified, if feasible. The mock-up booklet shall be prepared in typewritten form on 8-1/2 by 11 inch paper with stapled or GBC binding, depending on thickness and shall take the form and basic format of that specified in MIL-O-24312 and NAVSEA 0900-LP-098-7010. The mock-up booklet shall contain, but not necessarily be limited to, the following (see 6.2.2):

(a) Front matter
  (1) Cover.
  (2) Title page.
  (3) Forward.
  (4) Glossary of terms.
  (5) Table of contents.
  (6) List of figures.
  (7) Introduction
    a. Purpose and scope of mock-up booklet.
    b. Design philosophy and criteria employed.
    c. Design constraints imposed.

(b) General information chapter
  (1) Ship mission and tasks statement.
  (2) General characteristics.
  (3) Statement of operational concepts.
  (4) Brief description of Command and Control/Combat System.
  (5) Identification of Command and Control/Combat System spaces and stations.
  (6) Space or station relationship diagram.
  (7) Plan and profile views of ship with Command and Control/Combat System spaces or stations highlighted to show physical relationships within the ship.

(c) Information chapters (per space or station mocked-up)
  (1) General description of space or station.
  (2) Functional statement of space or station purpose.
  (3) Identification and description of functional areas within the space, inclusive of major equipment and facilities provided, operator and operator, and operator and machine interrelationship requirements.

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(4) Key plan of space or station showing physical location and immediate adjacency to other spaces.

(5) Detailed design rationale for space or station mocked-up inclusive of correlation matrices or other design work study techniques used in developing the detailed space/ station arrangement (see NAVSEA 0900-LP-098-7010).

(6) Plan view sketches (where necessary for clarity - elevation views) showing the overall arrangement of equipment, furniture and other facilities within the space or station, functional areas, and manned positions for each of the primary conditions of readiness (such as condition I, III and IV) and special evolutions for which there is a significant manning impact within the space (such as special sea detail, or vertical replenishment (VERTREP). Principal equipment and facilities are to be identified by generic title or, if readily understood, by brief nomenclature (such as AN/SPA-25 vice range azimuth indicator). The arrangement and identification of equipment located in overhead and bulkhead fiddleboards or panels shall be shown in detail views.

(7) Manning requirements matrix identifying operational stations within a space (subdivided by functional area, if applicable) and manning requirements for operational conditions of readiness I, III and IV and applicable special evolutions (such as special sea detail, VERTREP, or CONREP). The matrix shall contain symbology for station manning, such as: O-Optional, X-manned and WR, when required. For stations where the operator also mans a dedicated interior communications circuit in the performance of his duties, the circuit shall also be identified (such as Captain's Talker (JA)).

(8) Interior communications matrix identifying interior communications circuits and facilities available to each operational station within the space. This matrix shall reflect operator position, circuit designation (by circuit name and designation), E-Call and selector facilities, and interphone capabilities for sound powered telephone systems, Naval Tactical Data System (NTDS) interphone facilities (CK), ship service telephones, announcing system (such as, 1MC or 5MC), inter-communications systems (amplified voice)(such as, 21MC or 39MC), voice tubes, pneumatic tube facilities, and Interior Voice Communication System (IVCS) (or equal) capabilities.

(9) Exterior remote communications matrix identifying remote exterior communications circuits and facilities available at each operational station within the space. This matrix shall reflect operator position, circuit designation, channel selection capability and whether the circuit is at a remote individual terminal, incorporated within an NTDS console or subconsole, or a capability within IVCS.

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Additionally, normal mode of operation (such as, guard or monitor) shall be indicated for that position located. The matrix shall indicate send or receive terminal locations for teletype equipment (such as LINK 14) within Command and Control/Combat System spaces.

(10) For chapters or sections addressing message processing areas, a traffic flow sketch shall be included or superimposed on the plan view sketch.

3.6.2 Outline operational stations book. For designs which involve new concepts of operation or tactics and those designs which are complex in terms of the interaction of men, equipment and software on Outline Operational Stations Book (OSB) may be specified by NAVSEA and in accordance with the data ordering document specified (see 6.2.2). The purpose of the outline OSB is to allow for dynamic presentation of the mock-up and to amplify and describe the relationships between components of the shipboard operational systems, particularly the actions and interactions of operators with their equipment or operating positions. The outline OSB also serves as input to the development of a type I or type II OSB. The outline OSB shall take the form and format of that specified by MIL-O-24312 and NAVSEA 0900-LP-098-7010 and be published as either part 2 or volume II of the mock-up booklet. In instances where a type I or type II OSB has been prepared for the design being presented, a separate outline OSB shall not be prepared. The type I or type II OSB shall be available at the mock-up site, as directed by NAVSEA.

3.7 Disposition of mock-ups and models.

3.7.1 Type I and type III mock-ups. Type I and type III mock-ups, inclusive of models employed, shall become the property of NAVSEA.

3.7.2 Type II mock-ups. Type II mock-ups, constructed by the shipbuilder, shall be retained at the location where constructed until dismantling is authorized by NAVSEA. Type II mock-ups, constructed by a contractor other than the shipbuilder, shall be retained at the location where constructed until direction to ship the mock-up to some other location is received, or authorization to dismantle the mock-up is issued by NAVSEA. Authorization to dismantle the mock-up, or direction for its shipment to some other location, will be issued as soon as practicable after the mock-up has served its usefulness at the site where it was constructed (see 3.3.4(b)). The disposition of type II mock-ups, as stated above, is inclusive of all contractor (such as shipbuilder or other contractor) constructed models within the mock-up.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Inspection of packaging. Sample packages and packs and inspection of the packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Packaging.

5.1.1 Level C packing. Type I and type III (and type II, if applicable) mock-ups and models (see 3.7.1 and 3.7.2) shall be packed in containers acceptable to the common carrier, which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules or Regulations or other rules as applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order (see 6.2.1), marking of shipping containers shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The primary purpose of a mock-up, as a design tool, is to facilitate the evaluation of the physical design adequacy of the space mocked-up, inclusive of all its component parts. Secondary purposes are to determine equipment installation sequences, verification of foundation requirements and clearances, facilitate equipment and systems check-out, provide for crew team training in the shipboard operational environment, and orientation of shipyard installation personnel and other Government visitors.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of mock-up required (see 1.2.1).
(c) Class of models required (see 1.2.2), if other than as specified in 3.3.6 through 3.3.6.7.
(d) Spaces to be mocked-up (see 3.2.1).
(e) Scale of type I or type III mock-ups, if other than as specified in 3.3.2.1.
(f) Actual equipment, furniture and fixtures to be incorporated in type II mock-ups and their associated weights (see 3.3.4.1).
(g) Place of presentation (see 3.4.1 and 3.4.3).
(h) Distribution of negatives and prints as specified (see 3.5.1).
(i) Whether outline operational station book is required (see 3.6).
(j) Special marking, if required (see 5.2).

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.4.2.2	Milestone plan	UDI-A-20412	10.1
3.4.5.1 (a)	Notice of revision/ specification change notice	DI-E-1126	---
3.4.5.1 (b)	Drawings engineering and and associated lists	DI-E-7031	---
3.4.5.1 (c)	Photographs, vessel	UDI-A-23027	---
3.6.1 and 3.6.2	Book, operational stations	UDI-H-23353	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4 or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Security classification. Mock-ups, mock-up booklets, mock-up drawings, photographs of mock-ups and other mock-up related documentation will be unclassified, unless otherwise directed. Shipboard secure electrical system type II(O) and type II(F) models and mock-up spaces shall conform to the requirements of MIL-STD-1680 to the extent directed by NAVSEA.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Navy - SH
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-17779D(SH)	MOCK-UPS FOR SHIPBOARD COMMAND AND CONTROL, COMBAT SYSTEMS, AND COMMUNICATIONS STATIONS AND SPACES

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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